



ADAPTIVE CONTROL ALGORITHMS FOR LOAD TORQUE CHANGES IN ELECTRIC DRIVES

Mamazoirov Jahongir Mirzohid oğlu

Fergana State Technical University, Department
of Electrical Engineering Student of group 50-23EEE.
jahongirmirzo003@gmail.com

Mukhtorov Dilmurod Nu'monjon oğlu

Fergana State Technical University, Department
of Electrical Engineering, scientific degree
Senior Lecturer. PhD
dimajone0909@gmail.com
<https://doi.org/10.5281/zenodo.18051636>

Abstract

This thesis is devoted to the study of adaptive control algorithms for load torque changes in electric drives. The work shows that load torque changes affect rotor speed, energy consumption, and wear of mechanical parts, and traditional control systems cannot effectively respond to these changes. The thesis analyzes adaptive control, model-based predictive control, fuzzy logic, and Model Predictive Control (MPC) algorithms. The results show that adaptive control systems reduce rotor speed deviation by 20–40%, accelerate torque generation time by 2–5 times, and optimize energy consumption. This approach increases the efficiency of electric drives, reduces wear of mechanical parts, and ensures reliability in industrial systems.

Keywords: electric drive, load torque, adaptive control, MPC, fuzzy logic

Annotatsiya

Ushbu tezis elektr yuritmalarda yuklama momentining o'zgarishiga moslashuvchan boshqaruv algoritmlarini o'rganishga bag'ishlangan. Ishda yuklama momentining o'zgarishi rotor tezligi, energiya sarfi va mexanik qismlarning eskirishiga ta'sir qilishi, an'anaviy boshqaruv tizimlari esa bu o'zgarishlarga samarali javob bera olmasligi ko'rsatib o'tiladi. Tezisdan adaptiv boshqaruv, modelga asoslangan prognozli boshqaruv, fuzzy-logika va Model Predictive Control (MPC) algoritmlari tahlil qilinadi. Natijalar shuni ko'rsatadiki, moslashuvchan boshqaruv tizimlari rotor tezligining og'ishini 20–40% ga kamaytiradi, moment shakllanish vaqtini 2–5 baravar tezlashtiradi va energiya sarfini optimallashtiradi. Ushbu yondashuv elektr yuritmalarning samaradorligini oshiradi, mexanik qismlarning eskirishini kamaytiradi va sanoat tizimlarida ishonchlilikni ta'minlaydi.

Kalit so'zlar: elektr yuritma, yuklama momenti, moslashuvchan boshqaruv, MPC, fuzzy-logika





Аннотация

Этот тезис посвящена исследованию алгоритмов адаптивного управления изменением момента нагрузки в электроприводах. В работе показано, что изменение моментанагрузки влияет на скорость вращения ротора, энергопотребление и износ механических частей, а традиционные системы управления не способны эффективно реагировать на эти изменения. В диссертации анализируются адаптивное управление, предиктивное управление на основе моделей, нечеткая логика и алгоритмы предиктивного управления на основе моделей (MPC). Результаты показывают, что адаптивные системы управления снижают отклонение скорости вращения ротора на 20–40%, ускоряют время формирования момента в 2–5 раз и оптимизируют энергопотребление. Такой подход повышает эффективность электроприводов, снижает износ механических частей и обеспечивает надежность в промышленных системах.

Ключевые слова: электропривод, момент нагрузки, адаптивное управление, MPC, фuzzi-логика

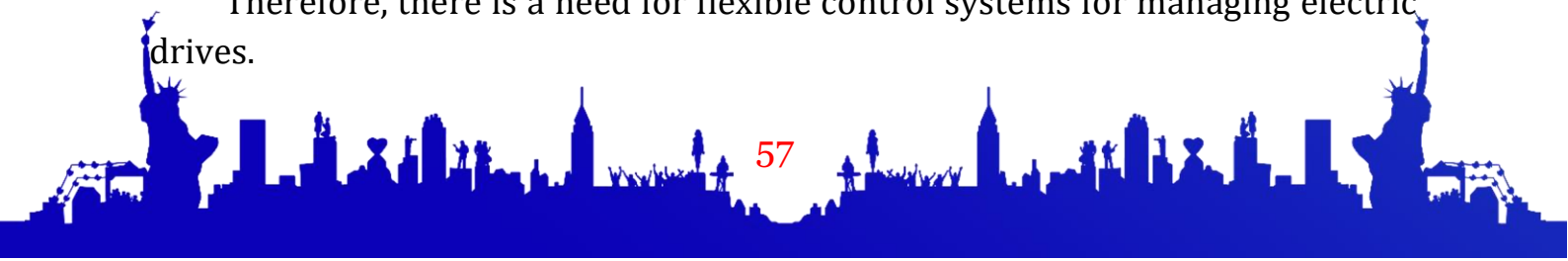
Login

In modern industrial processes, electric drives are one of the main elements that drive mechanical systems. In production, it is important to maintain a stable rotor speed, generate the required torque, and ensure energy efficiency[1,2]. In electric drives, changes in load torque have a significant impact on rotor speed, power consumption, heating rate, and wear of mechanical parts. Failure to control these processes leads to a decrease in production efficiency and the occurrence of failures in the short term. Therefore, the development of control algorithms that can adapt to changes in load torque in electric drives is an urgent issue[3].

Problem situation: Traditional control systems, including open-loop or simple PID controllers, cannot adapt to rapid and large changes in load torque. When the load increases, the rotor speed decreases, the required braking torque is not generated, and the system may lose its steady state. Conversely, when the load decreases, the excess torque increases energy consumption[2]. As a result:

- rotor speed deviation occurs;
- energy efficiency decreases;
- the level of vibration and heating increases;
- The wear of mechanical parts accelerates.

Therefore, there is a need for flexible control systems for managing electric drives.





Basic principles of adaptive control: The purpose of the adaptive control system is to recalculate the electric drive parameters in real time under the influence of load torque changes and maintain a stable torque. The following principles are used:

- determining the load state based on measurement signals;
- assessment and forecasting based on a system model;
- optimization of control parameters;
- generating compensation signals.

The most widely used algorithms are: adaptive control, model-based predictive control, fuzzy logic, Model Predictive Control (MPC)[1,3].

Adaptive control algorithms: An adaptive control system works by adapting to changes in parameters[1]. When the load torque changes, the system evaluates the parameters through identification blocks and automatically updates the regulator coefficients[2].

Model-based predictive control: In a model-based control system, a mathematical model of the electric drive is created[2]. Based on this model, the load torque is estimated and a compensation signal is generated[3].

Fuzzy logic algorithms: Fuzzy logic algorithms work effectively in conditions where uncertainty and nonlinear processes exist. The control signals change smoothly and increase the stability of the system[3].

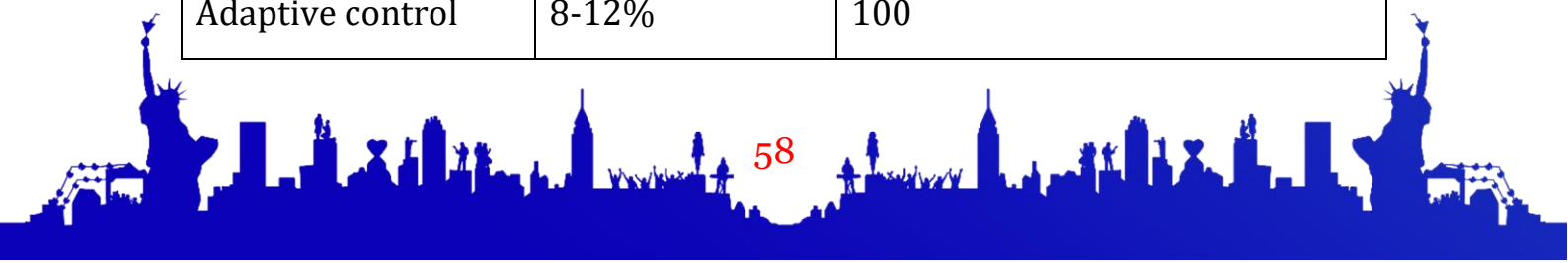
Model Predictive Control (MPC): The MPC algorithm predicts future states, is based on an optimality criterion, and controls boundary conditions[3].

Areas of application: Adaptive control algorithms can be applied in the following areas:

- conveyor and transport systems;
- crane and lifting mechanisms;
- robotic manipulators;
- pumps and fans;
- metalworking machines;
- compressor and turbomechanisms.

Results table

Algorithm type	Speed deviation	Torque generation time (ms)
Traditional	15-20%	200
Adaptive control	8-12%	100





Fuzzy logic	16-10%	120
MPC	4-7%	80

Conclusion

The thesis analyzes adaptive control algorithms for load torque changes in electric drives. The results show that adaptive control, fuzzy logic, and MPC algorithms reduce rotor speed deviation by 20–40%, accelerate torque generation time by 2–5 times, and increase energy efficiency. Adaptive control systems, compared to traditional PID regulators, reduce wear of mechanical parts and improve system stability. These algorithms can be effectively used in industrial devices, transport, and manipulator systems. In the future, their integration with artificial intelligence will allow for the creation of intelligent control systems.

Explanatory dictionary

Adaptive control – control that can adapt to changing parameters

Fuzzy logic – uncertainty-based control (or uncertainty-based control)

Model Predictive Control (MPC) – model-based predictive control

PID regulator – proportional-integral-derivative control device

Torque generation time – the time it takes for the torque to develop

References:

1. Mahkamov SJ Theory and control of electrical drives. – Tashkent: Fan Publishing House, 2020.
2. Leonhard W. Control of Electrical Drives. – Springer, 2019.
3. Smirnov A., Belov D. Model Predictive Control in Industrial Drive Systems. // International Journal of Electrical Engineering, 2022.

